

	Assignment No. 02 (Graded) Semester: Spring 2026 CS304P- Object Oriented Programming (Practical)	Total Marks: 20 Due Date: 28 June, 2026
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Uploading instructions:

- **Your assignment should be in .cpp files (Any other formats like scan images, doc/docx, PDF, zip, rar and bmp, etc. will not be accepted).**
- **Save your assignment with your ID (e.g. bc000000000.cpp).**
- **No assignment will be accepted through email.**

Rules for Marking:

It should be clear that your assignment will not get any credit if:

- The assignment is submitted after the due date.
- The submitted assignment does not open or the file is corrupted.
- Your assignment is copied from the internet, handouts, or any other student.
(Strict disciplinary action will be taken in this case).

Lectures Covered:

This assignment covers **Labs # 8 to 10**

Note: Submit your solutions in .cpp file only.

Assignment No. 2

Objective:

To understand and implement object-oriented concepts such as inheritance, abstract classes, virtual functions, and polymorphism in C++.

Learning Outcomes:

After completing this assignment, students will be able to implement the following concepts in C++:

- Inheritance
- Abstract and concrete classes
- Virtual functions
- Polymorphism

Problem Statement:

You are required to implement the **Tax Paying Management System** of FBR (Federal Board of Revenue) in Pakistan. The FBR treats differently the tax-payers by categorizing them into **filers** and **non-filers** while calculating the withholding tax.

Your **tax payment system** should be designed to support both types of taxpayers by structuring the classes in a hierarchy, where the **TaxPayer** class serves as an abstract base encapsulating shared behavior, and **FilerTaxPayer** and **NonFilerTaxPayer** serve as subclasses that inherit from it.

Each payer calculates and returns a **withholding tax while buying property** on the basis of following criteria.

The **FilerTaxPayer** is allowed to buy the property of any cost with tax-rate of 2 and half percent of total cost. $\text{WithHoldingTax} = 2.5\% \text{ of Total cost}$

The **NonFilerTaxPayer** is only allowed to buy the property of cost 30 lac or below with double tax-rate as the filer. **He/she cannot buy the property having cost above 30 lac.**

$\text{WithHoldingTax} = 5.0\% \text{ of Total cost}$, if eligible.

Detailed Description:

TaxPayer (**Abstract Base class**):

This class should include the following data members.

Data Members:

- PayerName
- PayerType
- tax-rate

Member functions:

- Parameterized constructor
- CalcWithHoldingTax() (**pure Virtual function**)
- DetailInfo() (**pure Virtual function**)
- Virtual Destructor

FilerTaxPayer (**Concrete class**):

Following are the members of this concrete class:

Data Members:

- ntn (National tax number)
- nFilerYears (number of years from which the payer is filer)

Member Functions:

- Parameterized constructor
- CalcWithHoldingTax() (override function)
- DetailInfo() (override function)
- Destructor

NonFilerTaxPayer (Concrete class):

Following are the members of this concrete class:

Data Members:

- IsEligible

Member Functions:

- Parameterized constructor
- CalcWithHoldingTax() (override function)
- DetailInfo() (override function)
- Destructor

Main Function Requirements:

- Declare a pointer variable of the base class (TaxPayer).
- Create the object of each sub-class dynamically and use polymorphism to call the CalcWithHoldingTax() and DetailInfo() functions of each sub-class.
- Properly delete the dynamically created objects at the end.

Note: PayerName, PayerType, tax-rate & ntn should be hard coded when creating the object in main() function while the PropertyCost should be entered by the user during the program execution. **Display the eligibility status of a non-filer also.**

Sample Output:

Output 1:

```
Enter the cost of property to buy: 2500000

***** FBR Tax Paying Management System *****

===== Filer Tax Payer =====
Payer Name      : Umar
Payer Type      : Filer
National Tax Number : 17301507
Property Cost: 2500000.000
Tax Rate        : 2.500%
Withholding Tax: 62500.000
Number of Years from which the Payer is filer: 3
*****

Destructor of FilerTaxPayer called for Umar

===== Non-Filer Tax Payer =====
Payer Name      : Ahmed
Payer Type      : Non-Filer
Eligible        : YES
Property Cost: 2500000.00
Tax Rate        : 5.00%
Withholding Tax: 125000.00
*****

Destructor of NonFilerTaxPayer called for Ahmed
```

Output 2:

```
Enter the cost of property to buy: 5000000

***** FBR Tax Paying Management System *****

===== Filer Tax Payer =====
Payer Name      : Umar
Payer Type      : Filer
National Tax Number : 17301507
Property Cost: 5000000.000
Tax Rate        : 2.500%
Withholding Tax: 125000.000
Number of Years from which the Payer is filer: 3
*****

Destructor of FilerTaxPayer called for Umar

===== Non-Filer Tax Payer =====
Payer Name      : Ahmed
Payer Type      : Non-Filer
Eligible        : NO
Property Cost: 5000000.00
Tax Rate        : 5.00%
Not eligible to buy property above 30 lac.
*****

Destructor of NonFilerTaxPayer called for Ahmed
```

Best of luck!

NOTE: Do not put any query on MDB about this assignment, if you have any query then email at cs304p@vu.edu.pk. Furthermore, if any student is found cheating by any other student or from online forums then he/she will be awarded ZERO right away and strict disciplinary action will be taken against the student.

Deadline: Your assignment must be uploaded/submitted on 28 June 2026.